

Supplemental Materials: Supporting Details and Secondary Tables

Yiliang Shao

S1 Problem and envelope illustrations

S2 Patent landscape search

A search of the US, EU, and WIPO patent corpora covered prior art on exact tower unit-mix stacking. The search keyword spread covered “unit mix,” “stacking plan,” “floor template,” “residential tower layout,” and combinations of those with “optimization” and “integer programming.” The most relevant patent located is US 11,727,173 B2, which discloses a pipeline for automated floor-plate layout including a *unit mix solver* component. The solver, as described in the patent’s preferred embodiments, is a delta-greedy heuristic operating on a single floor plate: it adds one unit per zone in order of largest deviation from the target mix, iterating until counts match. It is a greedy algorithm rather than an exact IP, and it operates per floor rather than across the building’s vertical stack with a building-wide K -cap.

No other patent in the searched corpora discloses tower-scale unit-mix stacking with a configuration-IP or any other exact method. No formal freedom-to-operate conclusion is drawn (that is a legal determination outside the methodological scope of the paper); the diligence is reported so that the absence of relevant exact-method prior art in the patent record is on the record.

S3 Consolidated formulations

For reference, the three solver formulations are consolidated below; the notation follows the main text.

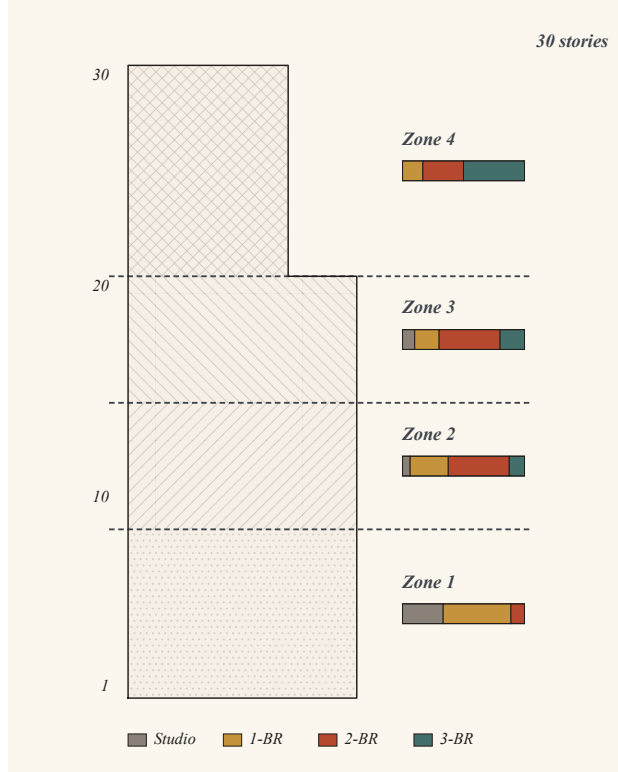


Figure S1: The unit-mix stacking problem: partition F floors into K contiguous zones, choose one unit-mix recipe per zone, and respect the program and plate-area budgets.

V1 (direct McCormick)

The full direct McCormick formulation (V 1), with all bilinearities linearized exactly: K zones indexed by z , types indexed by t , count grid $k \in \{0, 1, \dots, N_{\max}\}$ per (t, z) pair, indicator $\xi_{t,z,k}$ encoding $n_{t,z} = k$, and products $u_{t,z,k} = \xi_{t,z,k}a_{t,z}$ and $v_{t,z,k} = \xi_{t,z,k}f_z$. The V 1 baseline objective additionally carries a tie-breaking area-encouragement term with weight η (distinct from the proportional-mode band parameter β); this term is a benchmark-only artifact preserved from the original V 1 implementation and is absent from V 2 and V 2b. The feasible region is identical across V 1 and V 2 on every benchmark instance.

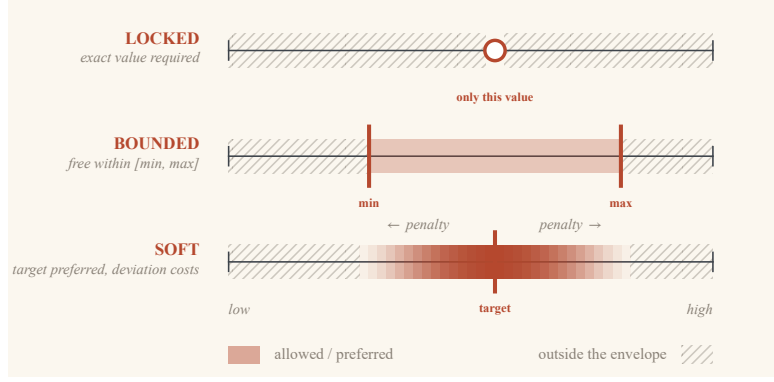


Figure S2: Every requirement is a knob in one of three states: locked (exact value required), bounded (free within $[\min, \max]$), or soft (target preferred, deviation penalized). The constraint envelope of the main text is the feasible region under the chosen knob states.

$$\begin{aligned}
\min \quad & \alpha \sum_t (d_t^+ + d_t^-) - \eta \sum_t a_t^* N_t + \gamma \sum_z u_z \\
\text{s.t.} \quad & \sum_z f_z = F, \\
& m u_z \leq f_z \leq F u_z, & \forall z \\
& \sum_k \xi_{t,z,k} = 1, \quad n_{t,z} = \sum_k k \xi_{t,z,k}, & \forall t, z \\
& a_t^{\min} \xi_{t,z,k} \leq u_{t,z,k} \leq a_t^{\max} \xi_{t,z,k}, & \forall t, z, k \\
& a_{t,z} - a_t^{\max}(1 - \xi_{t,z,k}) \leq u_{t,z,k} \leq a_{t,z} - a_t^{\min}(1 - \xi_{t,z,k}), & \forall t, z, k \\
& 0 \leq v_{t,z,k} \leq F \xi_{t,z,k}, & \forall t, z, k \\
& v_{t,z,k} \leq f_z, \quad v_{t,z,k} \geq f_z - F(1 - \xi_{t,z,k}), & \forall t, z, k \\
& A^{\text{lo}} u_z \leq \sum_{t,k} k u_{t,z,k} \leq A^{\text{hi}} u_z, & \forall z \\
& N_t = \sum_{z,k} k v_{t,z,k}, & \forall t \\
& \text{mode-specific count constraints on } N_t
\end{aligned}$$

with $f_z, n_{t,z}, v_{t,z,k}$ non-negative integer; $u_z, \xi_{t,z,k}$ binary; $a_{t,z}, u_{t,z,k}, d_t^\pm$ non-negative continuous within the stated bounds.

V2 (single-plate count-vector)

$$\begin{aligned}
\min \quad & \alpha \sum_t (d_t^+ + d_t^-) + \gamma \sum_p y_p \\
\text{s.t.} \quad & \sum_p x_p = F, \\
& m y_p \leq x_p \leq F y_p, & \forall p \\
& \sum_p y_p \leq K, \\
& A^{\text{lo}} y_p \leq \sum_t n_t^p a_{p,t}, & \forall p \\
& \sum_t n_t^p a_{p,t} \leq A^{\text{hi}} y_p + M_p(1 - y_p), & \forall p \\
& N_t = \sum_p n_t^p x_p, & \forall t \\
& \text{mode-specific constraints (below)} \\
& x_p \in \mathbb{Z}_+, y_p \in \{0, 1\}, a_{p,t} \in [a_t^{\min}, a_t^{\max}]
\end{aligned}$$

with $M_p = \sum_t n_t^p a_t^{\max}$ a template-tight constant fixed at enumeration time.

V2b (heterogeneous plates)

$$\begin{aligned}
\min \quad & \alpha \sum_t (d_t^+ + d_t^-) + \gamma \sum_{i \in \mathcal{I}} \sum_{p \in \mathcal{P}_i} y_{i,p} \\
\text{s.t.} \quad & \sum_{p \in \mathcal{P}_i} x_{i,p} = F_i, & \forall i \\
& m y_{i,p} \leq x_{i,p} \leq F_i y_{i,p}, & \forall i, p \\
& \sum_i \sum_p y_{i,p} \leq K, \\
& A_i^{\text{lo}} y_{i,p} \leq \sum_t n_t^{i,p} a_{i,p,t}, & \forall i, p \\
& \sum_t n_t^{i,p} a_{i,p,t} \leq A_i^{\text{hi}} y_{i,p} + M_{i,p}(1 - y_{i,p}), & \forall i, p \\
& N_t = \sum_i \sum_p n_t^{i,p} x_{i,p}, & \forall t \\
& \text{mode-specific constraints (below)}
\end{aligned}$$

Mode-specific constraint fragments

All unit types within an instance share one count mode. The formal fragments referenced from the main text are:

Proportional mode. With target proportions π_t satisfying $\sum_t \pi_t = 1$ and band parameter β , deviation pairs $d_t^+, d_t^- \geq 0$ enter with

$$N_t - \pi_t N = d_t^+ - d_t^-, \quad N = \sum_{s=1}^T N_s \quad \forall t, \quad (1)$$

and the hard band

$$\max(0, \pi_t - \beta) N \leq N_t \leq \min(1, \pi_t + \beta) N \quad \forall t. \quad (2)$$

The deviation pairs enter the objective as $\alpha \sum_t (d_t^+ + d_t^-)$.

Bounded-count mode. With per-type bounds $\underline{N}_t, \bar{N}_t \in \mathbb{Z}_+$ (the upper bound $\bar{N}_t$ may be omitted, in which case the upper inequality is dropped) and optional targets N_t^* ,

$$\underline{N}_t \leq N_t \leq \bar{N}_t \quad \forall t, \quad (3)$$

and for types with a target a soft deviation

$$N_t - N_t^* = d_t^+ - d_t^- \quad \forall t : N_t^* \text{ specified}. \quad (4)$$

Fixed-count mode. With per-type targets N_t^* ,

$$N_t = N_t^* \quad \forall t. \quad (5)$$

No deviation pairs are introduced; the soft term in the objective is absent.

S4 Supporting algorithms and case-study specification

Count-vector enumeration

For each plate class, enumeration performs a bounded depth-first recursion over unit types. At recursion depth j , the partial vector $(n_1, \dots, n_j)$ carries lower and upper area sums $S^{\min}$ and $S^{\max}$. A branch is pruned if even filling all remaining types at their maxima cannot reach the plate lower bound, or if the current lower area already exceeds the plate upper bound. A complete vector is retained when it is nonzero, respects the per-floor unit cap, and

satisfies the feasibility test in the main paper. Algorithm S1 gives the full recursion.

Algorithm S1 Count-vector enumeration for a single plate i with area band $[A^{\text{lo}}, A^{\text{hi}}]$ and per-type maxima $n_t^{\text{max}} = \min(N_{\text{max}}, \lfloor A^{\text{hi}}/a_t^{\text{min}} \rfloor)$. Plate-indexed quantities are fixed at the outer call and shown here without the i subscript for clarity.

```

1: procedure ENUMERATECV( $A^{\text{lo}}, A^{\text{hi}}, T, n^{\text{max}}, N_{\text{cap}}$ )
2:    $\mathcal{P} \leftarrow \emptyset$ 
3:    $n \leftarrow (0, \dots, 0)$ 
4:   RECURSE( $1, 0, 0$ )
5:   return  $\mathcal{P}$ 
6: end procedure
7: procedure RECURSE( $j, S^{\text{min}}, S^{\text{max}}$ )
8:   if  $j = T + 1$  then
9:     if  $\sum_t n_t > 0$  and  $\sum_t n_t \leq N_{\text{cap}}$  and  $S^{\text{max}} \geq A^{\text{lo}}$  and  $S^{\text{min}} \leq A^{\text{hi}}$ 
10:    then
11:       $\mathcal{P} \leftarrow \mathcal{P} \cup \{(n_1, \dots, n_T)\}$ 
12:    end if
13:    return
14:  end if
15:  for  $k = 0, 1, \dots, n_j^{\text{max}}$  do
16:     $S_{\text{new}}^{\text{min}} \leftarrow S^{\text{min}} + k \cdot a_j^{\text{min}}$ 
17:     $S_{\text{new}}^{\text{max}} \leftarrow S^{\text{max}} + k \cdot a_j^{\text{max}}$ 
18:     $P \leftarrow \sum_{\ell > j} n_{\ell}^{\text{max}} a_{\ell}^{\text{max}}$  ▷ residual upper bound
19:    if  $S_{\text{new}}^{\text{max}} + P < A^{\text{lo}}$  then
20:      continue ▷ underflow prune
21:    end if
22:    if  $S_{\text{new}}^{\text{min}} > A^{\text{hi}}$  then
23:      break ▷ overflow prune
24:    end if
25:     $n_j \leftarrow k$ 
26:    RECURSE( $j + 1, S_{\text{new}}^{\text{min}}, S_{\text{new}}^{\text{max}}$ )
27:  end for
28:   $n_j \leftarrow 0$ 
29: end procedure

```

Single-knob diagnostic

Starting from an infeasible instance, the diagnostic constructs one perturbed instance per candidate design knob: one plate-area tolerance, one unit-area tolerance, the compactness cap K , and, in bounded-count mode, one count window. Each perturbed instance changes exactly one knob by a fixed step and is solved under the same short budget. The first feasible probe identifies a sufficient single-knob relaxation; probes that remain infeasible identify non-binding knobs at that step. Algorithm S2 gives the full procedure.

Algorithm S2 Single-knob infeasibility diagnostic on an instance with parameters (τ, σ, K, N^*) and time budget Δt per probe.

```

1: procedure DIAGNOSE( $\Theta_0, \Delta\tau, \Delta\sigma, \Delta K, \Delta N$ )
2:   probes  $\leftarrow$  [ ]
3:   for plate  $i \in \mathcal{I}$  do
4:      $\Theta \leftarrow \Theta_0$  with  $\tau_i += \Delta\tau$ 
5:     probes.append((plate: $i$ .area_tol,  $\Delta\tau$ , SOLVE( $\Theta$ ,  $\Delta t$ )))
6:   end for
7:   for type  $t \in \{1, \dots, T\}$  do
8:      $\Theta \leftarrow \Theta_0$  with  $\sigma_t += \Delta\sigma$ 
9:     probes.append((unit: $t$ .area_tol,  $\Delta\sigma$ , SOLVE( $\Theta$ ,  $\Delta t$ )))
10:  end for
11:   $\Theta \leftarrow \Theta_0$  with  $K += \Delta K$ 
12:  probes.append((n_zones,  $\Delta K$ , SOLVE( $\Theta$ ,  $\Delta t$ )))
13:  if mode is bounded_count then
14:    for type  $t$  do
15:       $\Theta \leftarrow \Theta_0$  with  $\underline{N}_t -= \Delta N$ ,  $\overline{N}_t += \Delta N$ 
16:      probes.append((bounded: $t$ .window,  $\pm\Delta N$ , SOLVE( $\Theta$ ,  $\Delta t$ )))
17:    end for
18:  end if
19:  return probes
20: end procedure

```

Table S1: The synthetic case-study instance: plate classes (left) and unit program (right), $\sigma_t = 0.05$ for all types. The program sums to 63,484 m² against a plate envelope of 62,940 m² ($\sim 0.9\%$ over nominal, absorbed by the $\pm 5\%$ band); the studio count 58 is the Diophantine blocker at $K = 3$ in the main text.

plate label	A_i (m ²)	F_i (floors)	τ_i (baseline)
M (main)	1380	37	0.005
T (transition)	700	12	0.005
C (crown)	580	6	0.005
Total	—	55	—

type	a_t^* (m ²)	N_t^*
studio	40	58
1B	55	228
2B	78	264
2B+D	92	120
3B	118	144
Total	—	814

Instance tables

Solver-driven headroom curve

Synthetic case-study specification

The synthetic case-study instance is specified by the plate classes and unit types reported in the main paper, fixed-count mode, per-type tolerance $\sigma_t = 0.05$, baseline per-plate tolerance $\tau_i = 0.005$, minimum floors per zone $m = 2$, and $K \in \{3, 4, \dots, 11\}$. The script `experiments/case_study_ksweep.py` constructs the instance, runs the K-sweep and diagnostic, and writes `experiments/results/case_st`. The scripts `experiments/case_study_envelope.py`, `experiments/case_study_headroom.py`, and `experiments/case_study_k8_stacking.py` write the data underlying the case-study figures.

Table S2: The De Piek instance: rentable plate classes (left), traced from the architect’s published, scaled section, net of the vertical core and corridor; and the posed unit program (right), fixed counts, nominal areas NIA under the same net-area convention, $\sigma_t = 0.05$ for all types.

plate (zone)	A_i (m ² , net)	F_i (floors)
lower	940	3
mid	850	8
upper	640	8
Total	—	19

unit type	area (m ²)	count	area total (m ²)
studio	50	12	600
1B	68	33	2,244
2B	92	64	5,888
3B	120	40	4,800
PH	155	8	1,240
Total	—	157	14,772

S5 LP-relaxation tightness

The configuration LP relaxation of a Gilmore–Gomory-style formulation is known to be very strong on the classical cutting-stock problem (the IRUP/MIRUP property; Scheithauer and Terno 1995; Rietz and Scheithauer 2002). Tightness for the present formulation does not follow automatically: the y_p activation, the building-wide K -cap, and the mode-specific hard band interact in ways the classical formulation does not face.

Empirically, the relevant gap depends on the objective regime. In proportional mode with minimize-deviation the LP drives deviation to zero by fractional assignment while the MILP must absorb integer rounding, so the relative gap pins to 1.0 whenever the MILP objective is positive; the degeneracy lies in the metric, not the relaxation. In bounded mode with a maximize-units objective both objectives are positive and the gap measures relaxation quality. On the 30 solved cells of a 48-cell battery over plate classes, floors, unit types, and K -cap offsets, the maximum observed relative gap

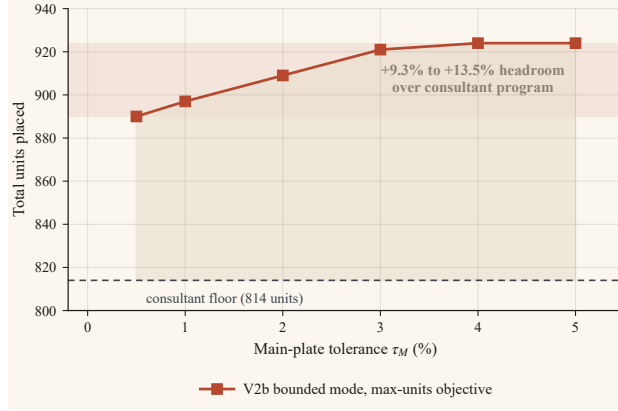


Figure S3: Solver-driven program headroom on the synthetic case-study instance at $K = 8$ (bounded-count mode, maximize-units objective): 890 units (+9.3% over the consultant program) at the baseline tolerance, saturating at 924 units (+13.5%).

is 2.78×10^{-9} , within solver feasibility tolerance; Gurobi 13.0.2 gives the same conclusion ($\max |g| = 1.52 \times 10^{-9}$). The LP relaxation is therefore empirically exact on this family, and no structural claim is made. Scripts `lp_tightness_battery.py`, `gurobi_lp_gap.py`; data `lp_gap*.csv`.

S6 Secondary experiment tables

Table S3: K-sweep of the synthetic case study at baseline $\tau_i = 0.005$ under HiGHS, 300 s per cell. Optimal[†] = primal feasible but dual gap not closed; Not Solved = no primal found.

K	status	zones used	solve time (s)	total units
3	Infeasible	—	44.1	—
4	Not Solved	—	301.2	—
5	Not Solved	—	301.4	—
6	Not Solved	—	301.3	—
7	Optimal [†]	7	301.1	814
8	Optimal [†]	7	301.2	814
9	Optimal [†]	6	301.3	814
10	Optimal [†]	6	301.4	814
11	Optimal [†]	6	301.4	814

Table S4: V 2b scaling diagonal moved from the main manuscript.

$ \mathcal{I} $	F	T	K	enum. time (s)	solve time (s)	total (s)	$\sum_i \mathcal{P}_i $
1	40	5	1	0.03	13.5	13.5	4362
2	40	4	2	0.005	1.2	1.2	557
3	40	4	3	0.005	3.0	3.0	821
4	40	4	4	0.008	5.9	5.9	1085
5	70	4	5	0.008	9.4	9.4	1349
2	40	6	6	0.113	152.3	152.4	22218

Table S5: Best-effort GA outputs on the case-study instance.

K	seed	wall (s)	type violation (units)	active templates	reported feasible
3	42	13.0	9	5	no
3	43	14.8	13	4	no
3	44	15.8	8	4	no
8	42	18.1	10	7	no
8	43	16.4	8	8	no
8	44	18.6	11	8	no

Table S6: Single-knob diagnostic probes at $K^* = 3$.

knob	step	status	solve time (s)
plate:M.area_tol	+0.50%	Infeasible	37.6
plate:T.area_tol	+0.50%	Infeasible	40.3
plate:C.area_tol	+0.50%	Infeasible	27.2
unit:studio.area_tol	+2.00%	Infeasible	21.7
unit:1B.area_tol	+2.00%	Infeasible	43.6
unit:2B.area_tol	+2.00%	Infeasible	45.8
unit:2B+D.area_tol	+2.00%	Infeasible	34.3
unit:3B.area_tol	+2.00%	Infeasible	38.3
n_zones	+1	Not Solved	181.4